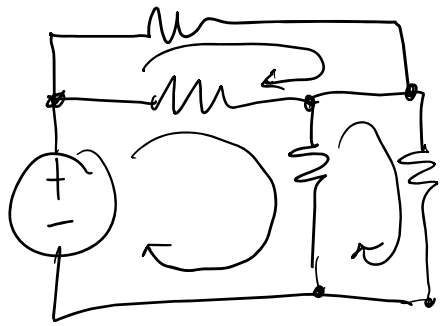
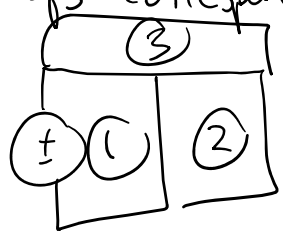
Lecture 2

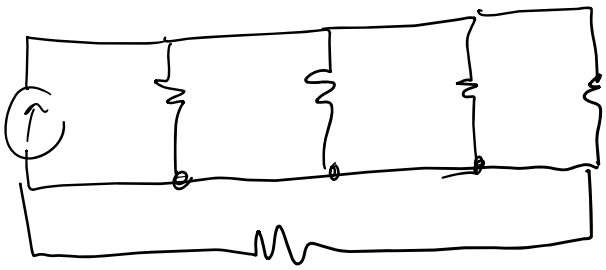
How to identify "Loops"



→ Loops correspond to "Boxes"

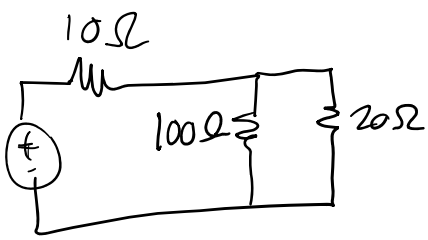


3-Boxes

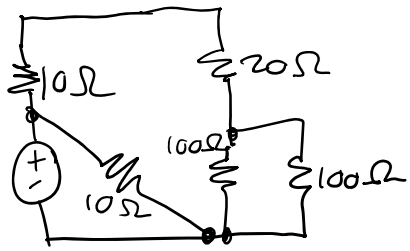


<u>Class activity</u>: find the number of loops and nodes in each circuit then; Reduce the Resistor using series and parallel.

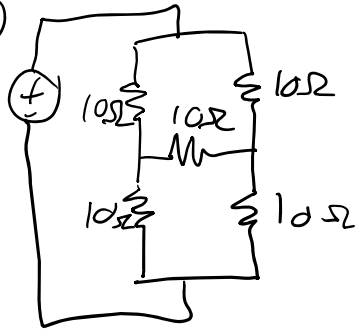
1.



2.

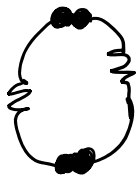


3.



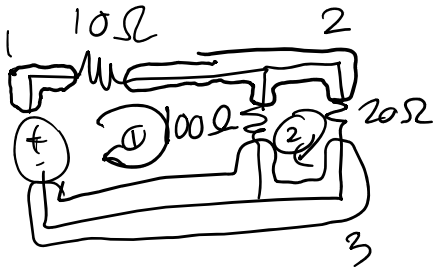
* series can only share 2-junctions

* Parallel must share both top junctions and both bottom junctions



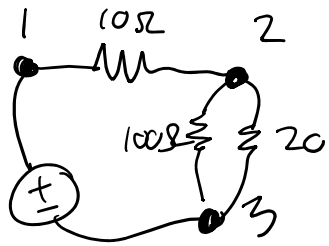
junction of element.

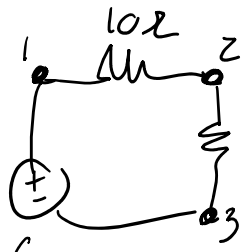
(1)



3 bubbles → each bubble is a node

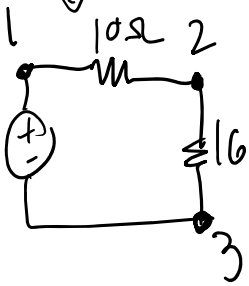
- 3 nodes.
- 2 Loops

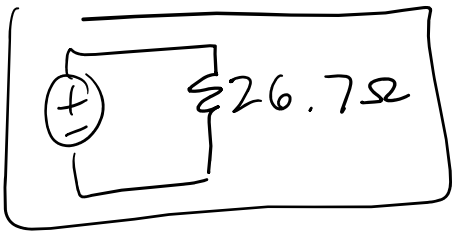




$$R_{\parallel} = \frac{1}{\left(\frac{1}{100}\right)+\left(\frac{1}{20}\right)} = \frac{1}{\left(\frac{20+100}{2000}\right)}$$

$$= \frac{1}{\left(\frac{120}{2000}\right)} = \frac{2000}{120} = 16.7$$





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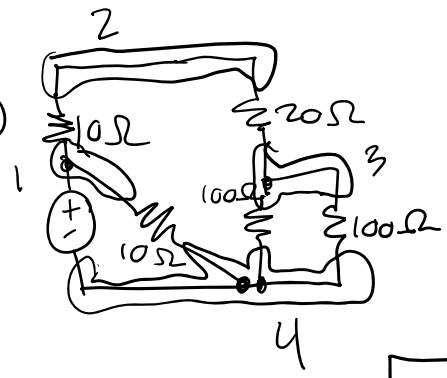
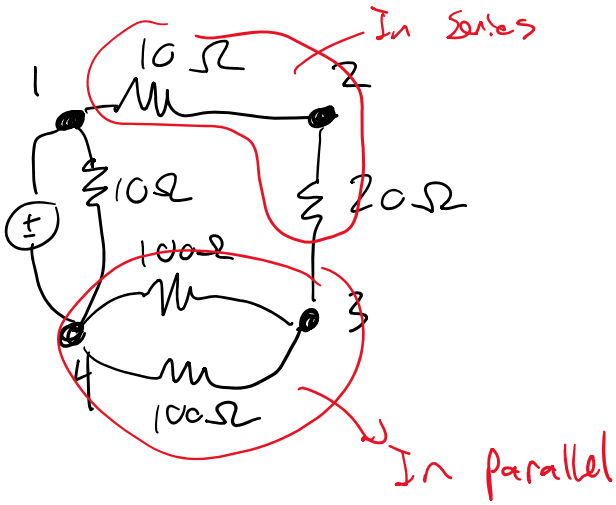
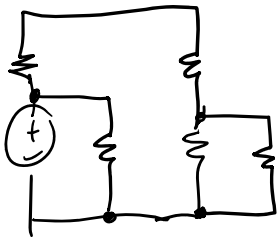
(2)

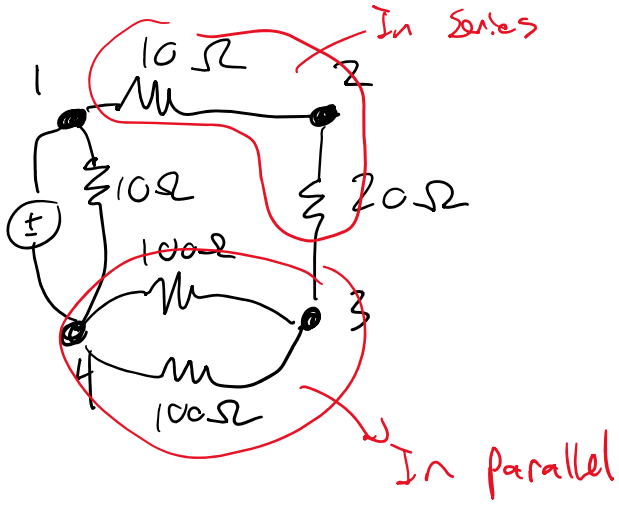




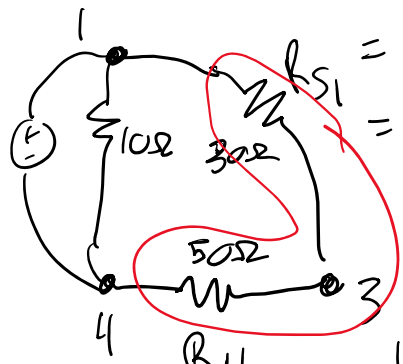
4-nodes

3-loops





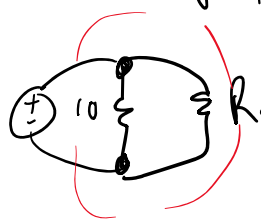
* Parallel connection
<u>Shunt</u> connection.



$R_{S_1} = 10\Omega + 20\Omega$
$= 30\Omega$

$R_{||_1} = \dfrac{1}{\left(\frac{1}{100}\right)+\left(\frac{1}{100}\right)} = \dfrac{1}{\left(\frac{2}{100}\right)} = \dfrac{100}{2}$

$R_{||_1} = 50\Omega$



$R_{S_2} = 30 + 50\Omega$
$= 80\Omega$

$R_{||_2} = \dfrac{1}{\left(\frac{1}{10}\right)+\left(\frac{1}{80}\right)} = \dfrac{1}{\frac{90}{800}} = \dfrac{800}{90}$

$$\boxed{R_{||_2} = 8.89\,\Omega}$$

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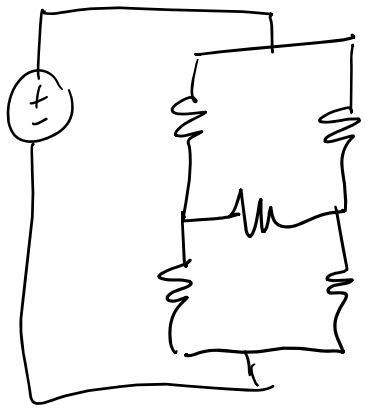
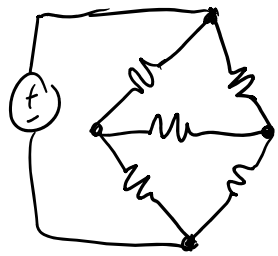
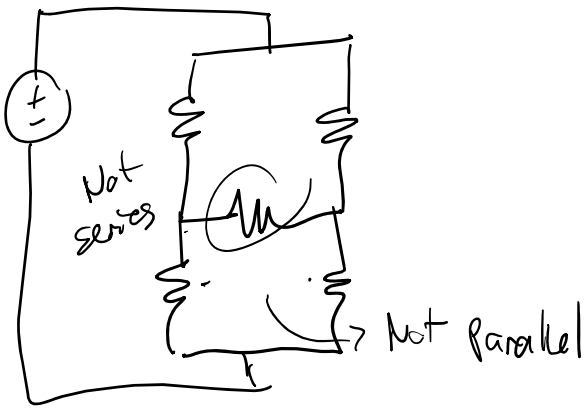
(3)



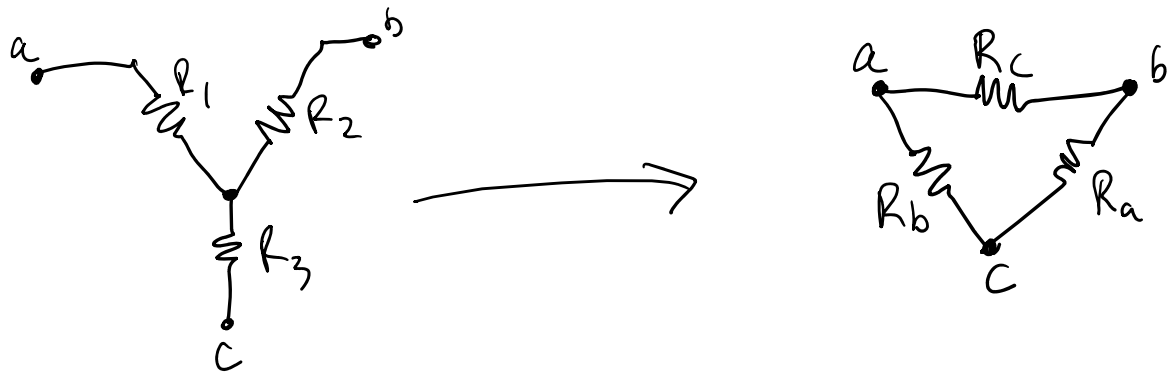
<u>Wheat stone Bridge</u>



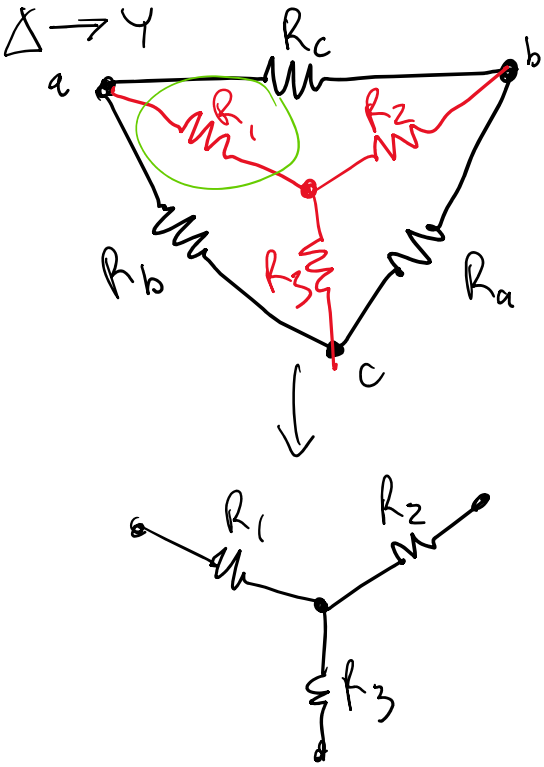


a, b, c are <u>nodes</u>

<u>wye - delta transformation</u>



$\Delta \rightarrow Y$

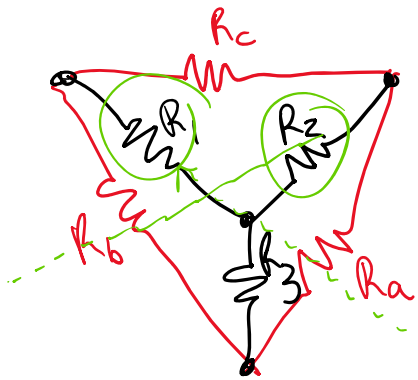


$$R_1 = \frac{R_c R_b}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

Y → Δ

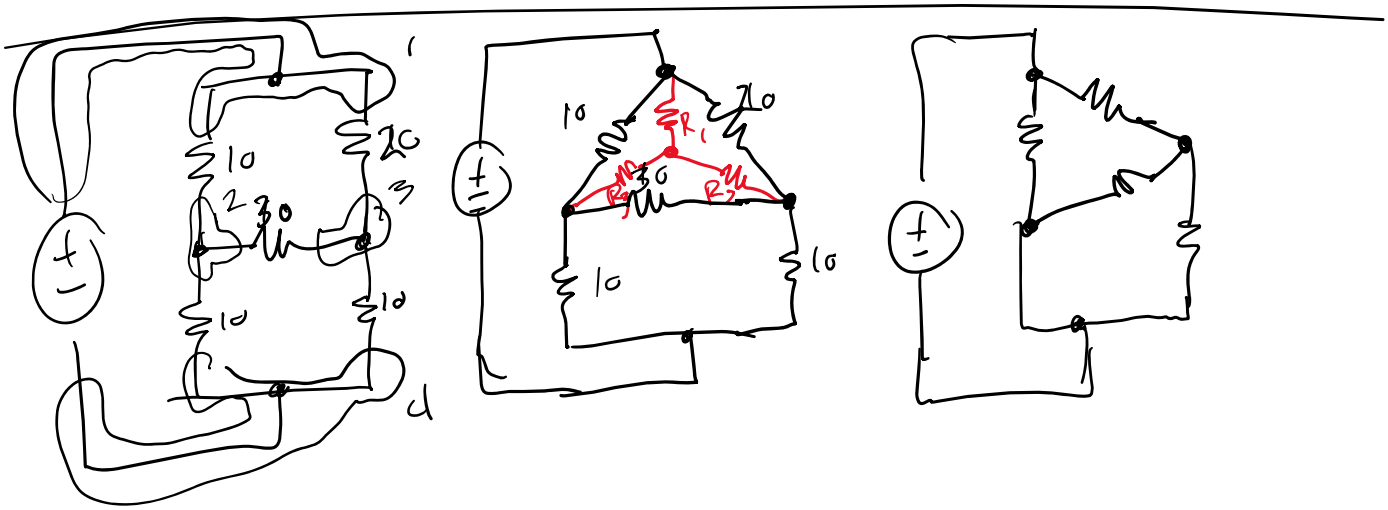


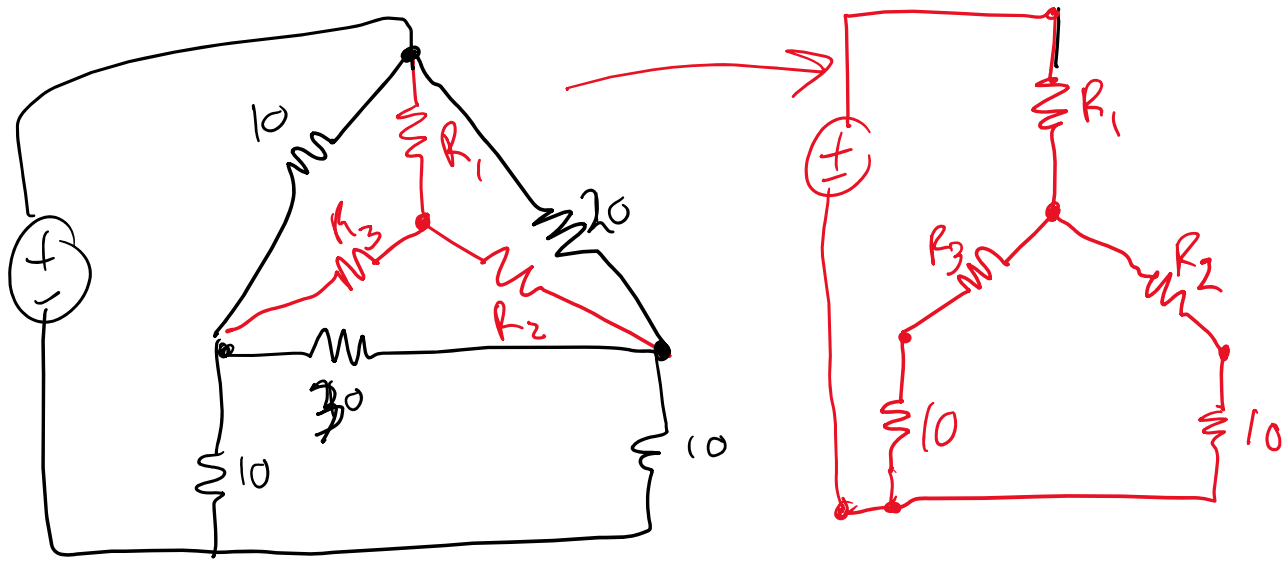
finding

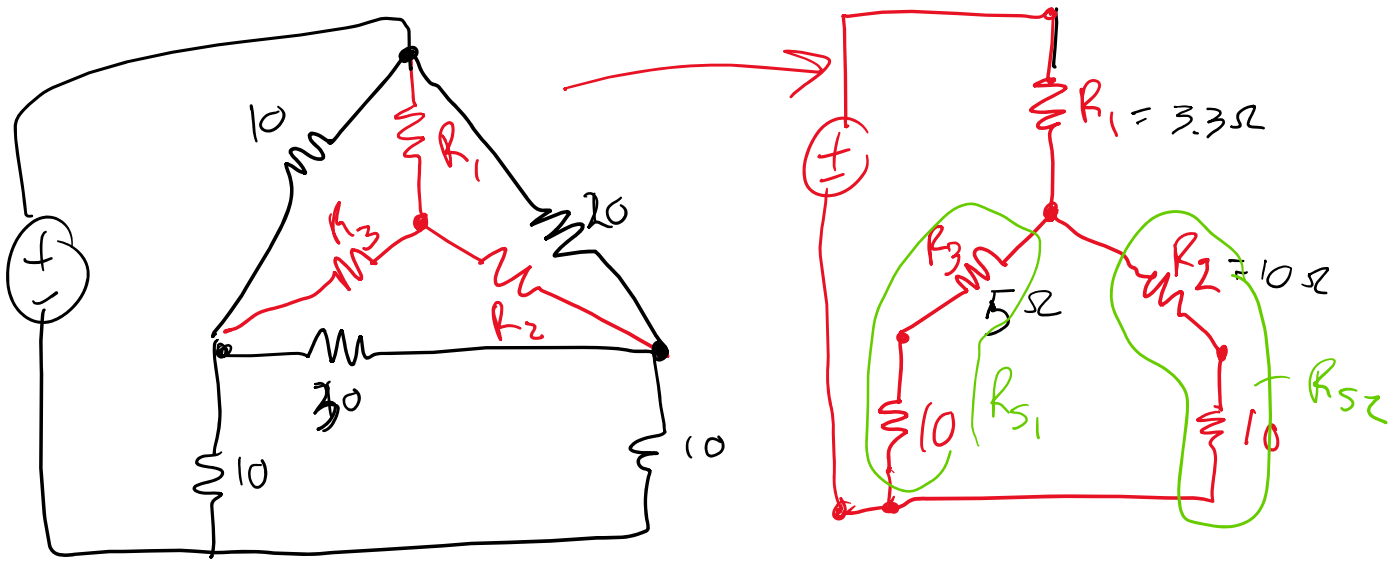
$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$



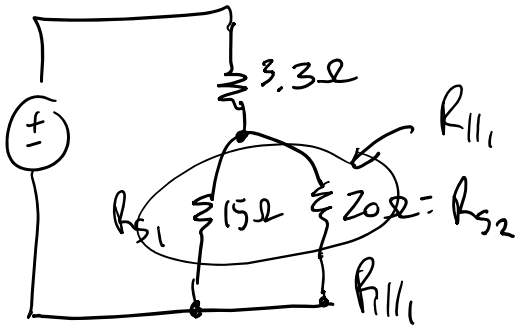




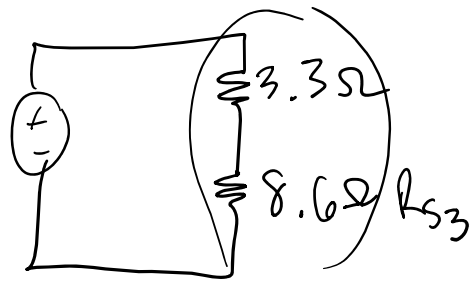
$$R_1 = \frac{(10)(20)}{10+20+30} = \frac{200}{60} = 3.3\Omega$$

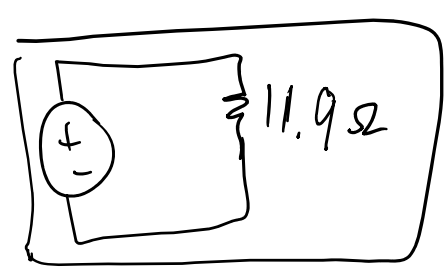
$$R_2 = \frac{(20)(30)}{10+20+30} = \frac{600}{60} = 10\Omega$$

$$R_3 = \frac{(10)(30)}{10+20+30} = \frac{300}{60} = 5\Omega$$



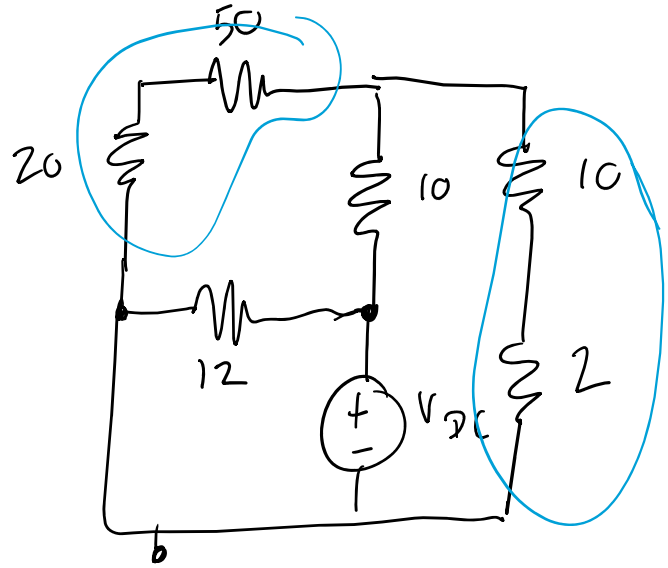
$$R_{\parallel_1} = \frac{1}{\left(\frac{1}{15}\right)+\left(\frac{1}{20}\right)} = 8.57\Omega$$





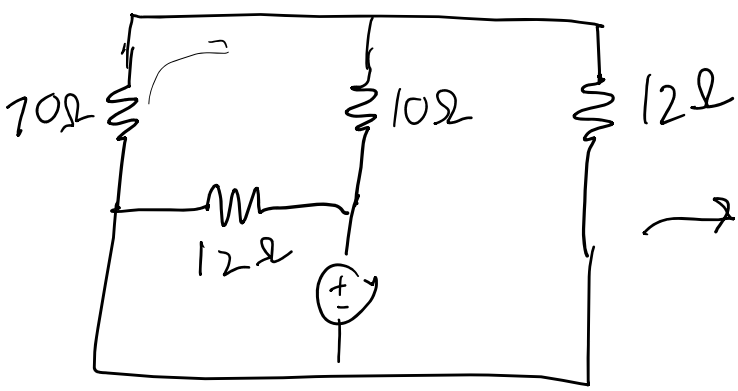
Exp 1

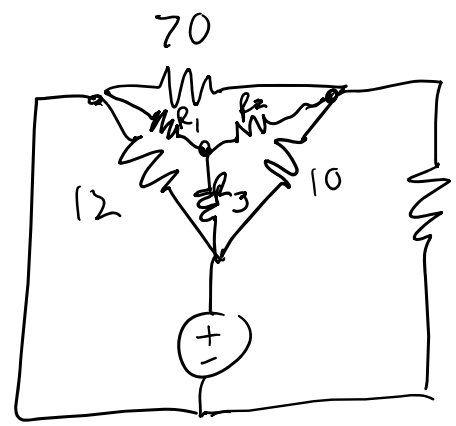
*Reduce!



• Look for series Parallel first then look for transformation.

1st Reduction



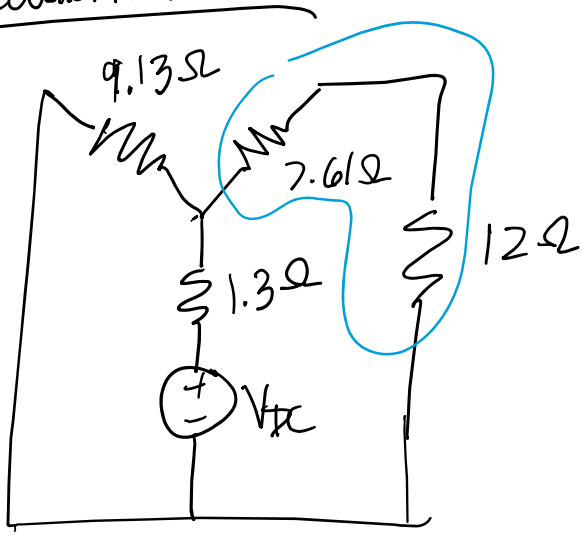


$$R_1 = \frac{12(70)}{(70+12+10)} = \frac{840}{92} = 9.13\Omega$$

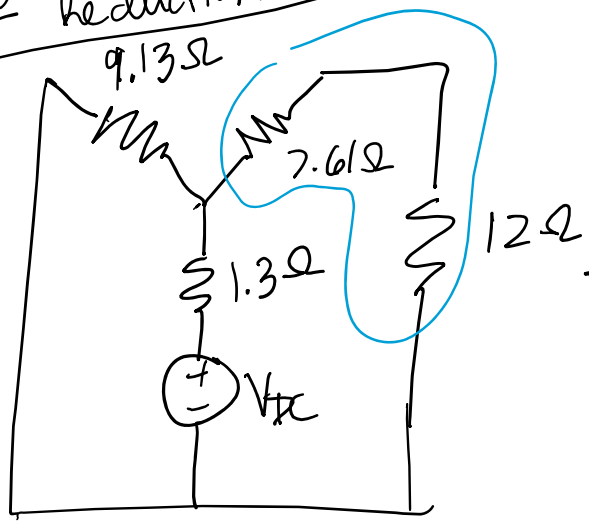
$$R_2 = \frac{10(70)}{70+12+10} = \frac{700}{92} = 7.61\Omega$$

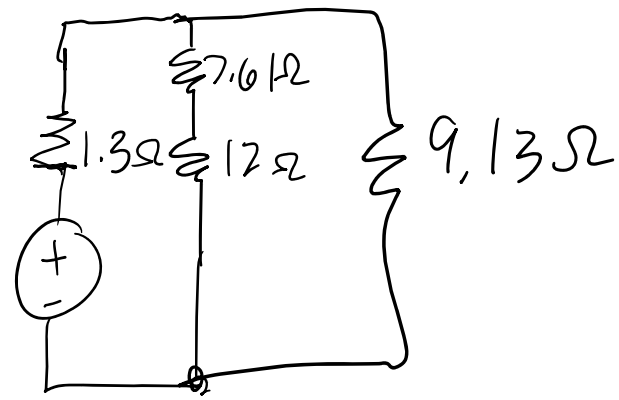
$$R_3 = \frac{12(10)}{70+12+10} = \frac{120}{92} = 1.3\Omega$$

Reduction 2



3rd Reduction

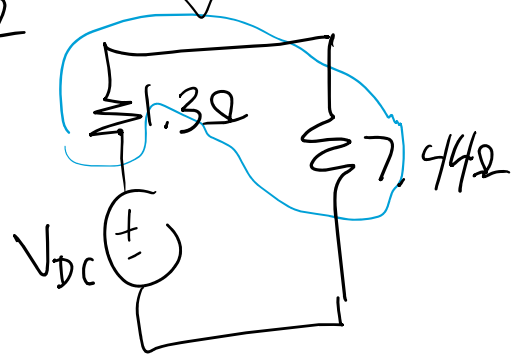
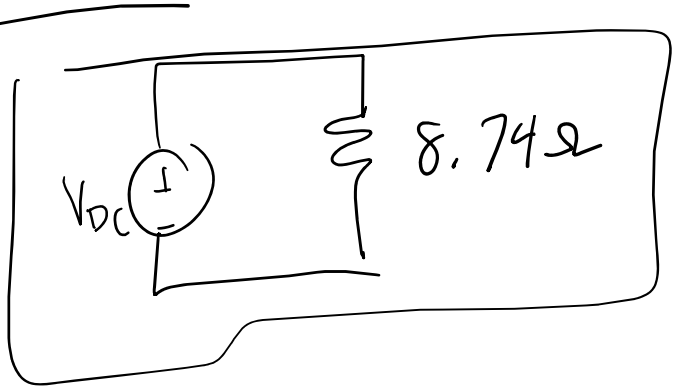




$$R_S = 7.61 + 12 = 19.61\Omega$$

$$R_P = \frac{1}{\left(\frac{1}{9.13}\right) + \left(\frac{1}{19.61}\right)}$$

$$R_P = 7.44\Omega$$



Last Reduction